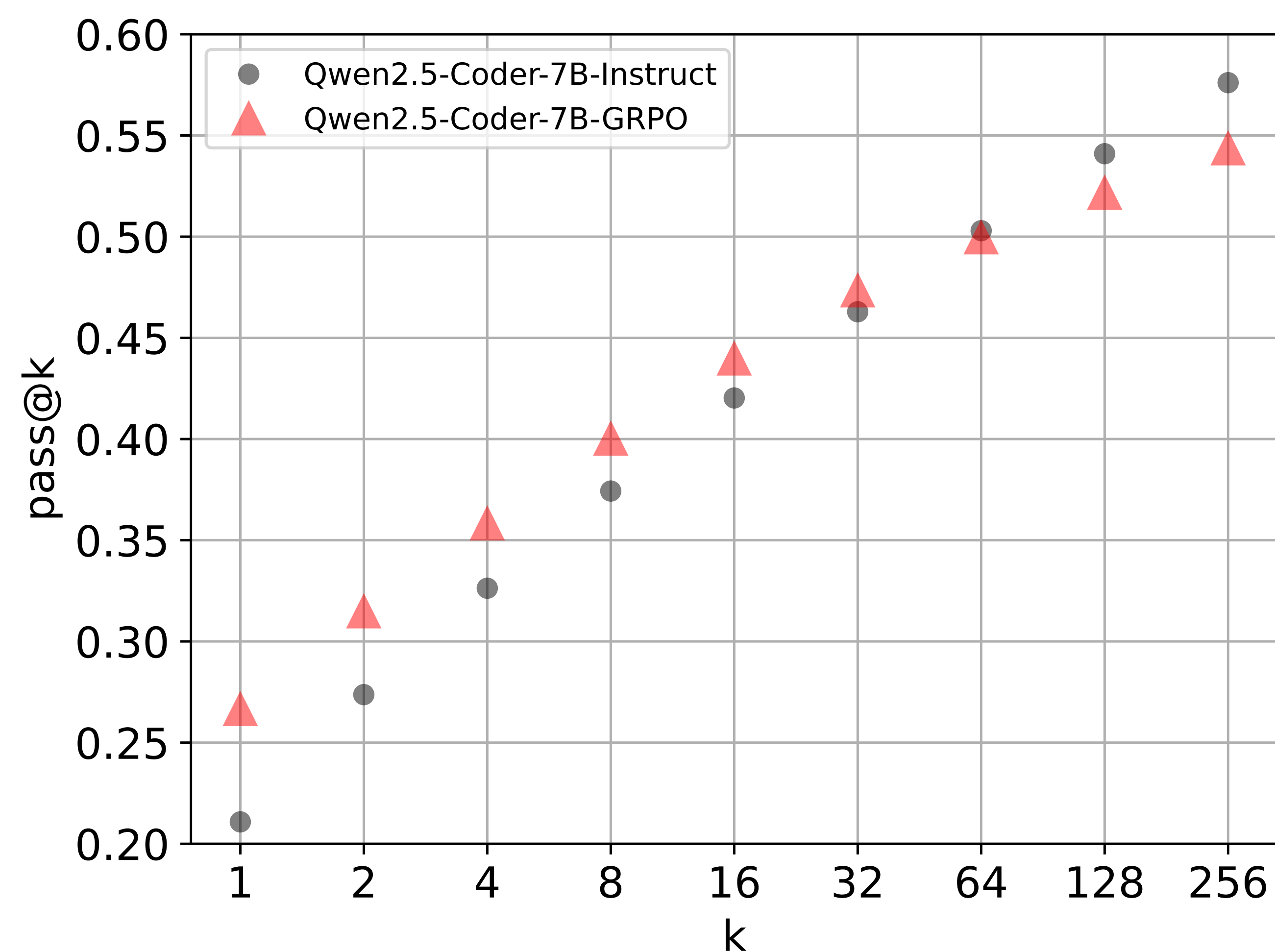


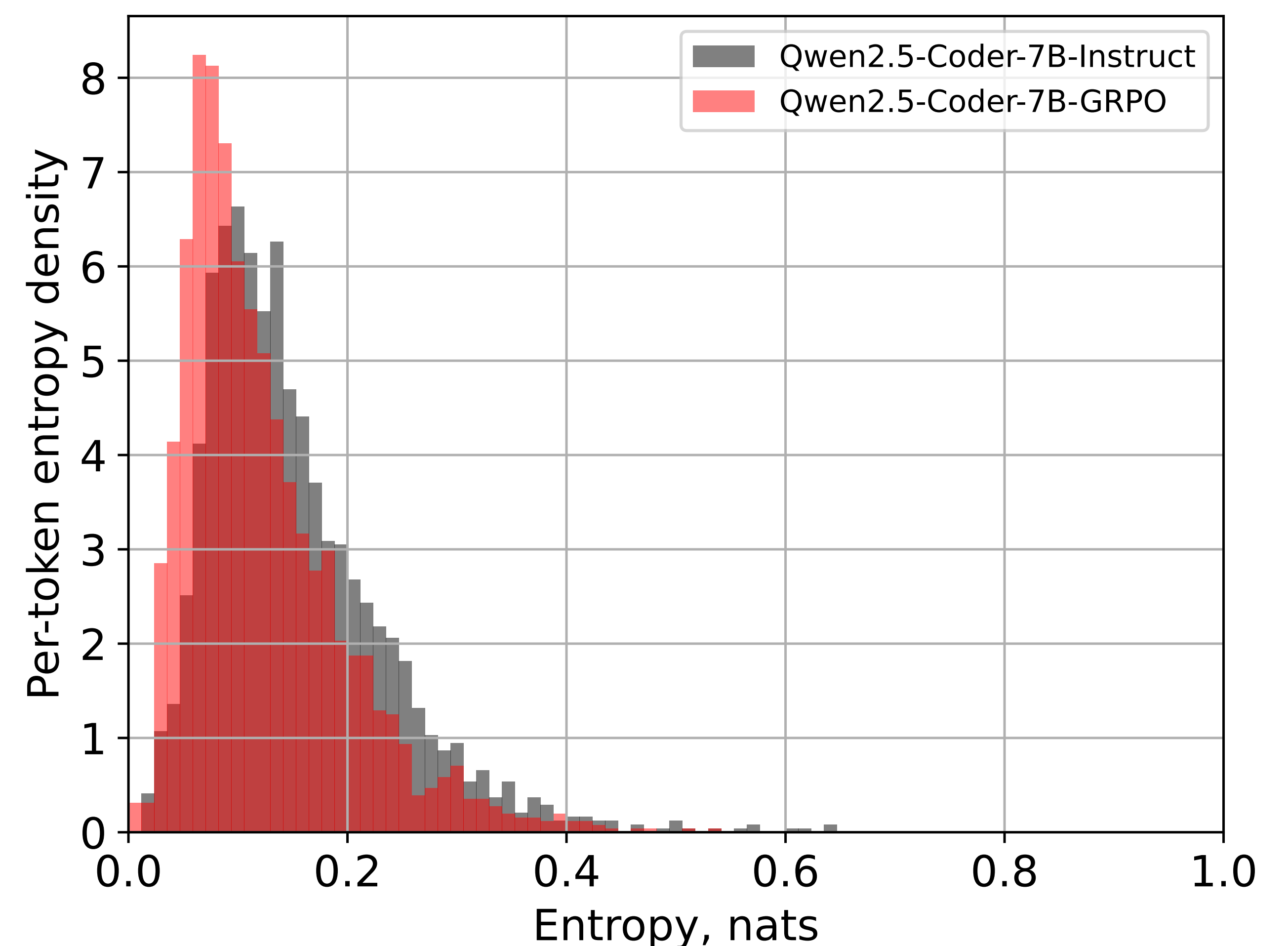
BEST OF N WORLDS

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pass@k of the base and GRPO fine-tuned models



Entropy distribution of the base and GRPO models



Overview

The application of Reinforcement Learning with Verifiable Rewards (RLVR) to mathematical and coding domains has demonstrated significant improvements in reasoning and problem-solving abilities of Large Language Models. Despite its success in single generation problem solving, the reinforcement learning fine-tuning process may harm the model’s exploration ability, as reflected in decreased diversity of generations and a resulting degradation of performance during Best-of-N sampling for large N values. In this work, we focus on optimizing the max@k metric, a continuous generalization of pass@k. We derive an unbiased on-policy gradient estimate for direct optimization of this metric. Furthermore, we extend our derivations to the off-policy updates, a common element in modern RLVR algorithms, that allows better sample efficiency.

Problems

The application of RLVR shows a significant improvement in performance in the single generation setting. Despite its success in a single generation setting the fine-tuned models suffer from:

- Decreased diversity of generations
- Lower scores for high values of k
- Optimization difficulties for the binary reward

Contributions

- We show that optimization of continuous reward is crucial for successful RLVR application.
- We show empirically the effect of RLVR on finetuned models: decreases diversity while increasing the confidence of its generations.
- We derive an unbiased estimate of —the generalization of for continuous rewards—gradients for on-policy and off-policy updates. This aligns the fine-tuning process with the inference strategy applied at the test time.
- We show that the proposed objective can efficiently optimize for higher k ’s while boosting lower k ’s as well, giving up to +3.5% increase in compared to baselines.

Core contribution

The core contribution of the paper is introduction of unbiased off-policy estimate for the optimization of Best-of-N sampling:

$$\frac{1}{\binom{n}{k}} \sum_{i=1}^n \left(\nabla_{\theta} \log \pi(y_i | x) \sum_{j=0}^n w'_{ij} r(y_j, x) \right).$$

$$w'_{ii} = \begin{cases} \binom{i-1}{k-1} (1 + \delta_i) + \binom{k-2}{i-2} \sum_{l < i} \delta_l, & i \geq k \\ 0, & i < k \end{cases},$$

$$w'_{ij} = \begin{cases} \binom{j-2}{k-2} (1 + \delta_i + \delta_j) + \binom{j-3}{k-3} \sum_{l < j, l \neq i} \delta_l, & j \geq k, j > i, \\ 0, & \text{otherwise} \end{cases}.$$

Results

Results for the binary vs. continuous optimization.

Setting	Code Contests		LiveCodeBench	
	pass@1	pass@128	pass@1	pass@128
Base model	0.211	0.541	0.211	0.510
Continuous reward	0.257	0.516	0.268	0.493
Binary reward	0.092	0.227	0.117	0.230

Results for max@k optimization for the proposed method and considered baselines

Method	CC	LCB	LB	CF	MBPP
Base model	<u>0.710</u>	<u>0.598</u>	0.241	0.226	0.619
BoN-max second	0.678	0.557	0.235	0.230	0.546
BoN-max mean	0.702	0.593	0.222	<u>0.242</u>	0.609
BoN LOO-1	0.647	0.530	0.227	0.200	0.484
BoN mean	0.693	0.579	0.280	0.229	<u>0.673</u>
Off-policy BoN (ours)	0.718*	0.616*	<u>0.278</u>	0.255*	0.710*